



LAB 03

I. System Requirements Context

The Equipment Borrowing Management System supports the process of requesting, approving, and tracking teaching equipment used by lecturers and students in a university. Currently, equipment is borrowed using paper forms or informal messages. This results in lost requests, double bookings, and a lack of usage reports. The system should digitalize the workflow as follows:

1: User Roles

- Borrower (Lecturer/Student):** searches for equipment, checks availability, submits borrow requests, and returns items.
- Approver (Department Head / Lab Technician):** reviews and approves borrow requests.
- Inventory Staff:** maintains equipment records, updates return status, and handles maintenance reports.
- Admin:** manages user accounts, system data, and reports.

2: Functional Requirements

- Users must **log in** to the system before submitting or approving requests.
- Borrowers can **search for equipment** and view real-time availability.
- Borrowers can **submit borrowing requests** with purpose, date/time, and quantity.
- Approvers can **approve or reject** borrow requests.
- Borrowers and staff can **view request status** (Pending / Approved / Rejected / Returned).
- The system sends **reminders for due dates** automatically.
- Inventory Staff can **record returns** and **update equipment availability**.
- Admin can **generate reports** on borrowing frequency, late returns, and utilization.

3: Non-Functional Requirements

- System must respond within **3 seconds** per transaction.
- Access is **web-based**, available to all users within the campus network.
- Data must be consistent and validated before updates.

4: Students will use this requirement context to identify:

- External entities (Borrower, Approver, Inventory Staff, Admin, Notification Service, etc.)
- Processes (Handle Requests, Approve Request, Update Inventory, etc.)
- Data flows (Request Form, Approval Notice, Borrow Slip, etc.)
- Data stores (Equipment, Borrow Records, Users, etc.)

5: These elements will be represented through DFDs and the Data Dictionary.

II. Tasks

1: Task 1 – Context Diagram

a. Identify system boundary and external entities (Borrower, Approver, Inventory Dept, etc.).

(1) System Boundary

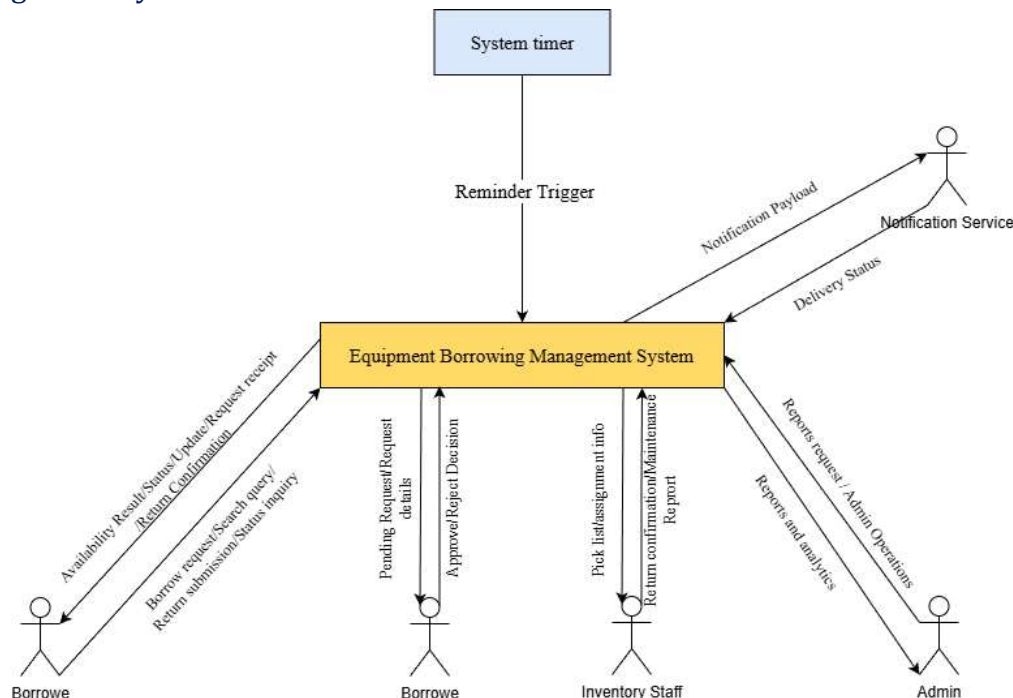
Equipment Borrowing Management System (EBMS)

Digitizes the process of requesting, approving, tracking, and returning teaching equipment at universities.

(2) External Entities

External Entity	Description / Role
Borrower (Lecturer / Student)	Searches for equipment, submits loan requests, views status, returns items
Approver (Department Head / Lab Technician)	Reviews and approves or rejects borrowing requests
Inventory Staff	Records returned items, updates availability and maintenance reports
Admin	Manages system users, equipment data, and reports
Notification Service	Sends email/app notifications and reminders
System Timer (Scheduler)	Automatically triggers due-date reminders and overdue checks

b. Draw the Context Diagram with one process representing the Equipment Borrowing Management System.



c. Major Data Inflows / Outflows

External Entity	Input to System	Output from System
Borrower	Borrow Request, Search Query, Return Submission, Status Inquiry	Request Receipt, Availability Result, Status Update, Return Confirmation
Approver	Approval / Reject Decision	Pending Request List, Request Details
Inventory Staff	Return Confirmation, Maintenance Report	Picklist / Assignment
Admin	Report Requests, Admin Configuration	Reports & Analytics
Notification Service	Delivery Status	Notification Payload
System Timer	Reminder Trigger	N/A

d. Check the balance with the next level.

2: Task 2 – Level-0 DFD

a. Processes

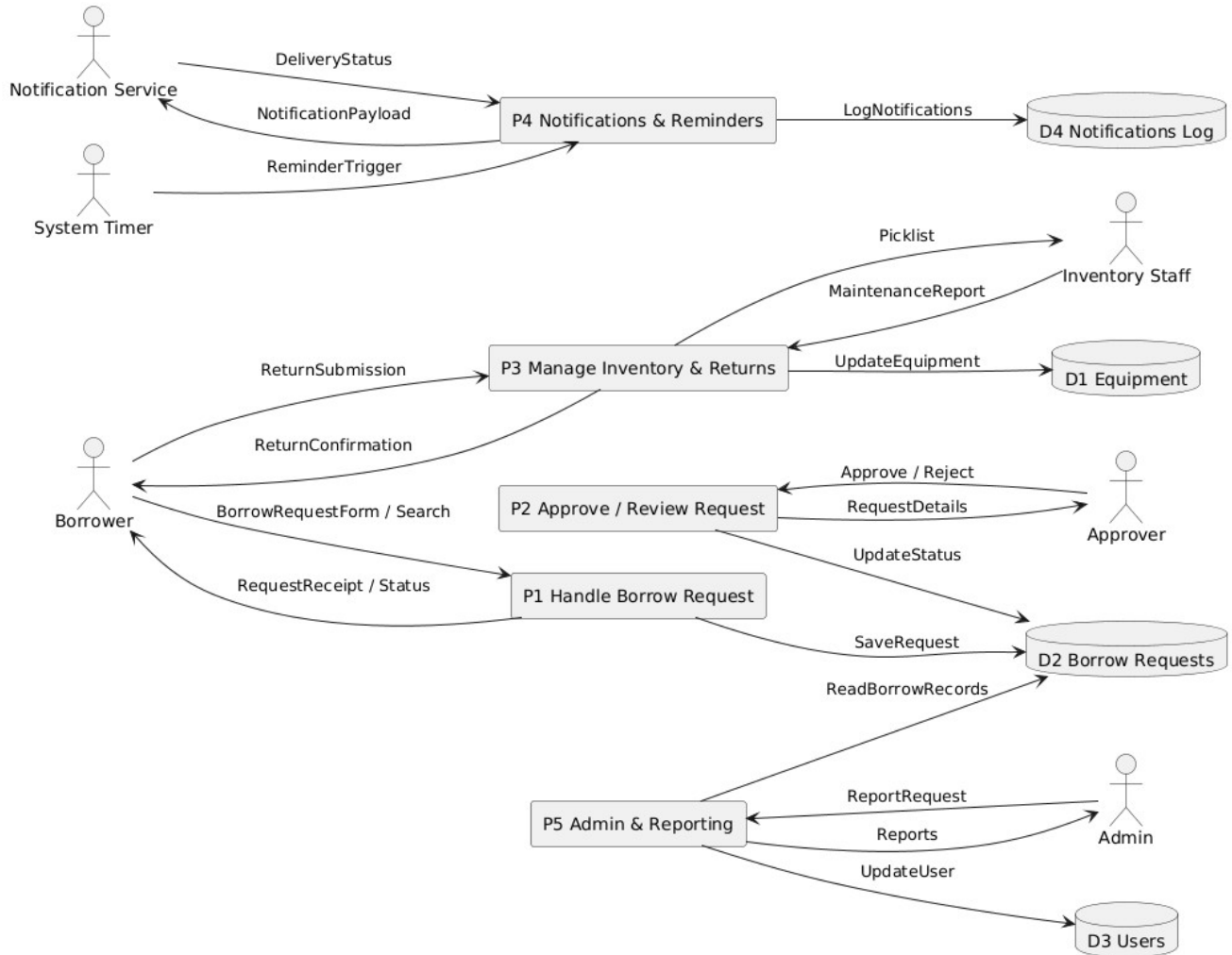
- P1 – Handle Borrow Request
- P2 – Approve / Review Request
- P3 – Manage Inventory & Returns
- P4 – Notifications & Reminders
- P5 – Admin & Reporting

b. Data Stores

- D1 – Equipment
- D2 – Borrow Requests
- D3 – Users
- D4 – Notifications Log

c. Level 0 DFD

Level-0 DFD — Equipment Borrowing Management System



3: Task 3 – Data Dictionary

(a) A. Data Elements (DE)

ID	Element Name	Description	Type / Format	Domain / Constraint	Example
DE01	RequestID	Unique borrow request ID	String(20)	NOT NULL	REQ-2025-001
DE02	DeviceID	Equipment identifier	String(20)	NOT NULL	EQP-101
DE03	UserID	Unique user ID	String(20)	NOT NULL	ST12345
DE04	StaffID	Approver/Inventory Staff ID	String(20)	FK	ST67890
DE05	BorrowDate	Borrow date	Date	≥ today	45971
DE06	ReturnDate	Planned return date	Date	≥ BorrowDate	45976

DE07	Quantity	Number of items	Integer	≤ stock	2
DE08	Status	Request status	Enum	Pending / Approved / Returned	Approved
DE09	Purpose	Borrowing purpose	String(100)	Optional	Classroom lecture
DE10	MaintenanceStatus	Equipment condition	Enum	Available / Under Maintenance	Available
DE11	ReportType	Report category	Enum	Borrowing / Utilization / Late	Borrowing
DE12	ReportPeriod	Reporting period	String(20)	Optional	Q1-2025
DE13	NotificationID	Notification identifier	String(20)	Auto-generated	NTF-101
DE14	MessageContent	Notification message	Text	Required	"Your request was approved."
DE15	DeliveryStatus	Notification result	Enum	Delivered / Pending / Failed	Delivered

(b) B. Data Structures (DS)

ID	Structure	Components (DE)	Mandatory	Notes
DS01	BorrowRequest	DE01, DE02, DE03, DE05, DE06, DE07, DE08, DE09	Yes	Borrower's submission
DS02	ApprovalRecord	DE01, DE04, DE08	Yes	Approval decision
DS03	EquipmentRecord	DE02, DE10	Yes	Equipment details
DS04	ReportRequest	DE11, DE12	Optional	Requested by Admin
DS05	NotificationMessage	DE13, DE14, DE15	Yes	Notification log

(c) C. Data Stores (D)

ID	Name	Description	Key Field	Updated by (Process)
D1	Equipment	Equipment details and status	DeviceID	P3 Manage Inventory

D2	Borrow Requests	Borrow requests and status	RequestID	P1, P2, P3
D3	Users	User accounts and roles	UserID	P5
D4	Notifications Log	Notification details	NotificationID	P4

(d) D. Data Flows (F)

ID	Name	Source	Destination	Structure	Purpose
F01	Borrow Request Form	Borrower	P1	DS01	Submit request
F02	Request Receipt	P1	Borrower	DS01	Confirm submission
F03	Return Submission	Borrower	P3	DS01	Submit return
F04	Approval Decision	Approver	P2	DS02	Approve/reject request
F05	Equipment Update	Inventory Staff	P3	DS03	Update status
F06	Report Request	Admin	P5	DS04	Request system report
F07	Notification Payload	P4	Notification Service	DS05	Send notification
F08	Delivery Status	Notification Service	P4	DS05	Confirm delivery
F09	Reminder Trigger	System Timer	P4	N/A	Start notification

(e) E. Processes (P)

ID	Process Name	Description	Inputs	Outputs	Trigger
P1	Handle Borrow Request	Validate and record borrower requests	F01	F02, D2	Borrower submits
P2	Approve / Review Request	Approver reviews and decides	F04	D2, F07	Approver action
P3	Manage Inventory & Returns	Update equipment and returns	F03, F05	D1, D2	Return or update
P4	Notifications & Reminders	Send notifications and reminders	F09, D2	F07, F08	Timer event

P5	Admin & Reporting	Manage users and generate reports	F06	D3, D4	Admin request
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4: Notes for Students

- Every flow, store, and process listed here must also appear in your DFD.**
- Use consistent names and IDs (e.g., P1 ↔ Handle Borrow Request).
- The **Data Dictionary is the textual counterpart** of your DFD.
- You may add more elements or structures if your team's scenario includes special features (e.g., reminders, maintenance reports, notifications).

III. Submission

The Equipment Borrowing Management System (EBMS) provides a structured digital workflow for borrowing, approving, and returning equipment within the university.

Through the Context Diagram, Level-0 DFD, and Data Dictionary, the system's data movements and responsibilities among entities were clearly defined.

This model supports real-time tracking, eliminates double bookings, automates reminders, and enhances transparency for both staff and students.

It demonstrates the importance of proper system analysis and design in creating efficient, user-centered management systems.